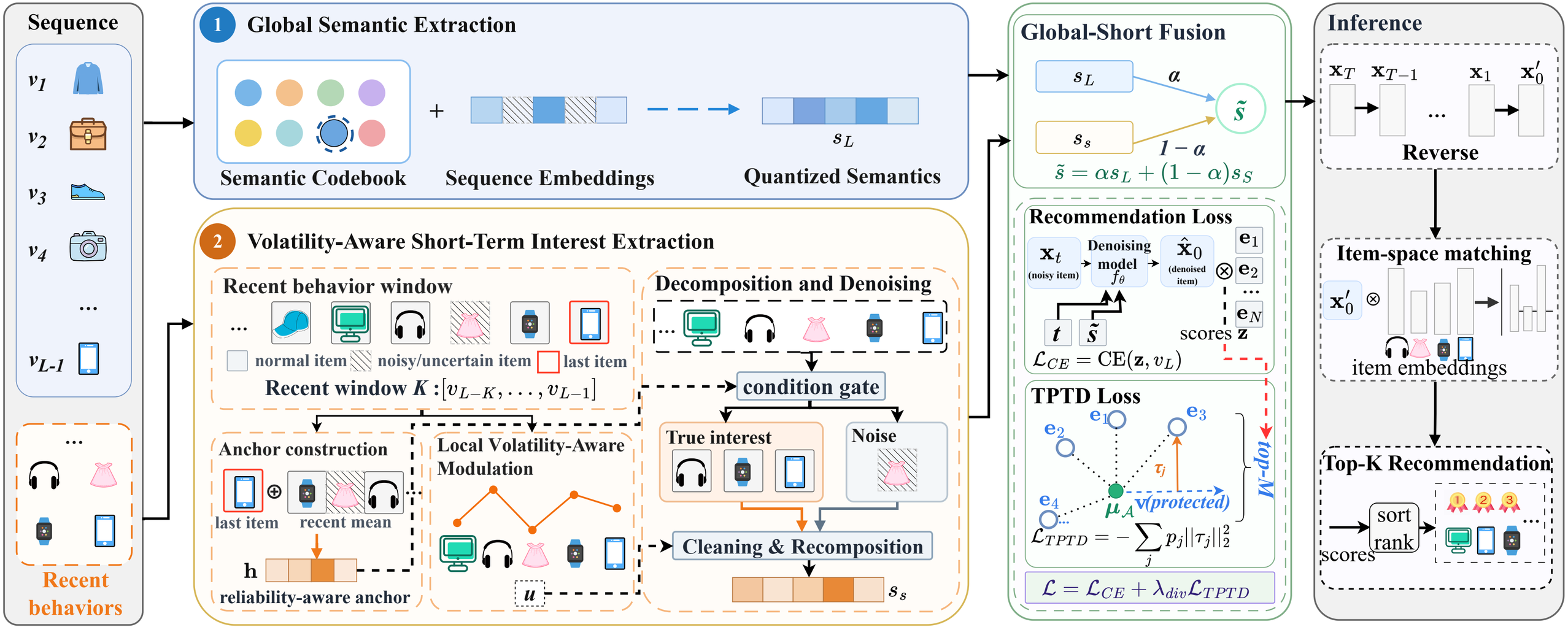

Sequence
v_1
v_2
v_3
v_4
...
v_{L-1}
...
Recent behaviors
1 Global Semantic Extraction
Semantic Codebook
+
Sequence Embeddings
s_L
Quantized Semantics
2 Volatility-Aware Short-Term Interest Extraction
Recent behavior window
normal item
noisy/uncertain item
last item
Recent window K : $[v_{L-K}, \ldots, v_{L-1}]$
Anchor construction
last item
recent mean
h
reliability-aware anchor
Local Volatility-Aware Modulation
u
Decomposition and Denoising
condition gate
True interest
Noise
Cleaning & Recomposition
s_s
Global-Short Fusion
s_L
α
s_s
$1-\alpha$
$\tilde{s}$
$\tilde{s} = \alpha s_L + (1-\alpha) s_S$
Recommendation Loss
$\mathbf{x}_t$ (noisy item)
Denoising model f_θ
$\hat{\mathbf{x}}_0$ (denoised item)
$\mathbf{e}_1$
$\mathbf{e}_2$
...
$\mathbf{e}_N$
scores $\mathbf{z}$
t
$\tilde{s}$
$\mathcal{L}_{CE} = \mathrm{CE}(\mathbf{z}, v_L)$
TPTD Loss
$\mathbf{e}_1$
$\mathbf{e}_2$
$\mathbf{e}_3$
$\mathbf{e}_4$
τ_j
μ_A
v(protected)
top-M
$\mathcal{L}_{TPTD} = -\sum_j p_j ||\tau_j||_2^2$
$\mathcal{L} = \mathcal{L}_{CE} + \lambda_{div} \mathcal{L}_{TPTD}$
Inference
$\mathbf{x}_T$
$\mathbf{x}_{T-1}$
...
$\mathbf{x}_1$
$\mathbf{x}'_0$
Reverse
Item-space matching
$\mathbf{x}'_0$
item embeddings
Top-K Recommendation
scores
sort
rank